Annotated Bibliography

Raelyn Brooks

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References

[1] Albert-László Barabási and Márton Pósfai. *Network Science*. Cambridge University Press, 2016.

This textbook establishes the fundamental mathematical and topological principles of network science, exploring graph structures, connectivity, and complex systems. In this project, it provides the theoretical framework for representing mass spectrometry feature data as graph networks, where molecular features act as nodes and cosine similarity scores function as weighted edges.

[2] Roberto J. Bayardo, Yiming Ma, and Ramakrishnan Srikant. Scaling up all pairs similarity search. In *Proceedings of the 16th International Conference on World Wide Web*, pages 131–140, 2007.

This paper presents high-performance algorithms for computing all-pairs similarity across large sets of sparse, high-dimensional vectors. It offers algorithmic strategies relevant to optimizing backend cosine similarity matrix calculations for mass spectral datasets.

[3] Vincent Blondel, Jean-Loup Guillaume, Renaud Lambiotte, and Etienne Lefebvre. Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008:P10008, 10 2008.

This paper introduces the Louvain community detection algorithm for efficiently identifying clustered substructures within large networks. In this project, it serves as the algorithmic basis for automatically grouping related molecular features and chemical families within the interactive network visualization.

[4] Mathieu Jacomy, Tommaso Venturini, Sebastien Heymann, and Mathieu Bastian. Forceatlas2, a continuous graph layout algorithm for handy network visualization designed for the Gephi software. *PLOS ONE*, 9(6):e98679, 2014.

This paper details the ForceAtlas2 continuous force-directed layout algorithm, which positions nodes based on network connectivity to reveal structural clusters and hubs. It provides a methodological reference for the graph rendering and visualization mechanics used to organize chemical families and unknown transformation products in the software interface.

[5] Amy J. Ko, Robin Abraham, Laura Beckwith, Alan Blackwell, Margaret Burnett, Martin Erwig, Chris Scaffidi, Joseph Lawrance, Henry Lieberman, Brad Myers, Mary Beth Rosson, Gregg Rothermel, Mary Shaw, and Susan Wiedenbeck. The state of the art in end-user software engineering. *ACM Comput. Surv.*, 43(3), April 2011.

This paper examines software engineering principles tailored to end-user developers and domain experts. It supports the project's philosophy of designing transparent, localized computational tools that empower domain scientists to execute complex data workflows without relying on black-box platforms.

[6] Panos Louridas. Review of network science by albert-làaszlò barabàasi. *SIGACT News*, 49(2):10–13, June 2018.

This review of Barabási's foundational work highlights the interdisciplinary evolution of network science, bridging computer science, mathematics, and domain-specific applications. It provides helpful context for the thesis literature review, emphasizing how graph-theoretic tools can be successfully adapted to non-traditional computational domains like analytical chemistry.

[7] Yu A. Malkov and D. A. Yashunin. Efficient and robust approximate nearest neighbor search using hierarchical navigable small world graphs. *IEEE Trans. Pattern Anal. Mach. Intell.*, 42(4):824–836, April 2020.

This paper introduces Hierarchical Navigable Small World (HNSW) graphs for efficient approximate nearest neighbor search in high-dimensional vector spaces. In this project, it provides the algorithmic foundation for optimizing high-dimensional spectral vector comparisons, enabling rapid cosine similarity calculations and scalable pairwise distance indexing across large environmental LC-HRMS datasets.

[8] Robert Pienta, Acar Tamersoy, Hanghang Tong, Alex Endert, and Duen Horng (Polo) Chau. Interactive querying over large network data: Scalability, visualization, and interaction design. In *Companion Proceedings of the 20th International Conference on Intelligent User Interfaces*, IUI '15 Companion, page 61–64, New York, NY, USA, 2015. Association for Computing Machinery.

This paper explores scalable graph visualization, data mining, and interactive design principles for navigating complex network structures. It directly supports the user interface and interaction design goals of the custom software, informing how chemists can interactively query, filter, and inspect node metadata without needing rigid query languages.

[9] Greg Wilson, Jennifer Bryan, Karen Cranston, Justin Kitzes, Lex Nederbragt, and Tracy K. Teal. Good enough practices in scientific computing. *PLOS Computational Biology*, 13(6):e1005510, 2017.

This article outlines practical software engineering guidelines for data management, version control, and code organization tailored to computational research. It supports the project's core philosophy of developing transparent, reproducible, and sustainable localized software workflows designed specifically for domain experts.